

WEB PROGRAMMING PROJECT REPORT

E-Commerce Platform for an Online Enoteca

Student Information

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Course: Web Programming

Project Type: WP9 – E-commerce Platform

Academic Year: 2025/2026

1. Introduction

The purpose of this project is the development of a complete e-commerce website dedicated to the online sale of wines. The idea behind the project is to simulate a real Italian Enoteca that allows customers to browse products, create an account, log in, add products to a shopping cart, and manage purchases through a user-friendly web interface.

The project was developed following the requirements specified in the course guidelines and combines both frontend and backend web technologies. The application provides authentication features, database interaction, product management, and an administration area.

2. Project Motivation

The wine industry is an important sector of the Italian economy, and many local wine shops require an online presence to reach a wider audience. The objective of this project is to create a modern web platform that enables customers to discover and purchase wines online.

The project aims to reproduce the essential functionalities of a real e-commerce website while applying the concepts studied during the Web Programming course.

3. Technologies Used

The following technologies were used during the development of the project:

Frontend Technologies

HTML5

Used to create the structure of all web pages.

CSS3

Used to style the website and improve the user experience through responsive and modern layouts.

JavaScript

Used to enhance interactivity and improve client-side functionality.

Backend Technologies

PHP

Used for server-side programming and business logic implementation.

MySQL

Used as the relational database management system to store user and product information.

XAMPP

Used as the local development environment including Apache and MySQL services.

Development Tools

Visual Studio Code

Used as the main code editor.

Git

Used for version control.

GitHub

Used for source code hosting and project management.

4. System Architecture

The application follows a client-server architecture.

The client interacts with the website through a web browser.

The server processes requests using PHP scripts and communicates with the MySQL database.

Main components include:

- User Authentication System
- Product Catalog
- Shopping Cart
- Checkout Module
- Administration Dashboard
- Database Management Layer

5. Database Design

The database stores all information necessary for the application.

Main tables include:

Users Table

Stores:

- User ID
- Username
- Email
- Password

Products Table

Stores:

- Product ID
- Product Name
- Description
- Price
- Product Image

Cart Information

Stores products selected by users before checkout.

The database communication is handled through PHP and SQL queries.

6. Implemented Features

The following functionalities were implemented:

User Registration

Users can create a new account through the registration page.

Features

- Input validation
- Secure data storage
- Database insertion

User Login

Registered users can access their accounts.

Features

- Authentication system
- Session management
- Login validation

Product Catalog

The website displays available wines.

Features

- Product image display
- Product description
- Product price information

Shopping Cart

Users can add products to a cart before purchasing.

Features

- Add to cart
- Remove from cart
- Quantity management
- Total price calculation

Checkout Page

Users can review selected products before completing an order.

Administration Dashboard

An administrator can manage products.

Features

- Add Product
- Edit Product
- Delete Product
- View Product List

This functionality satisfies the requirement of having an administrative interface.

7. Website Structure

The project is organized into multiple modules:

Authentication Module

- login.php
- register.php
- logout.php

Product Module

- product.php

Shopping Module

- cart.php
- checkout.php

Administration Module

- dashboard.php
- add_product.php
- edit_product.php
- delete_product.php

Configuration Module

- database.php

8. Development Decisions

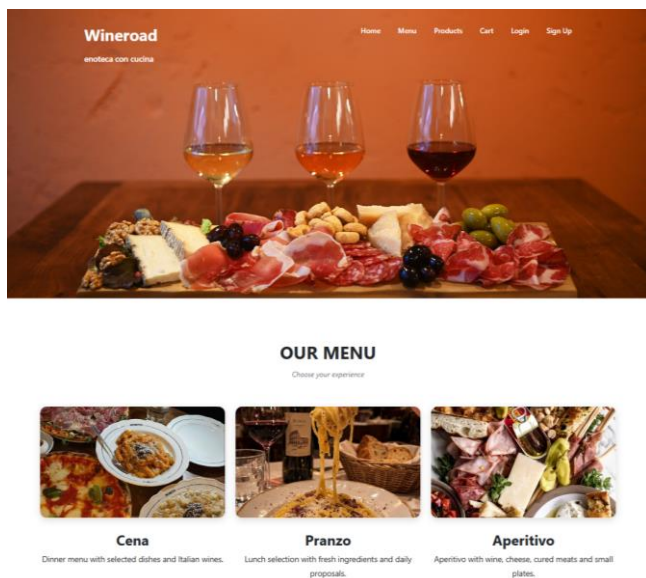
Several design decisions were taken during development:

- Use of PHP because it integrates easily with MySQL.
- Use of sessions to maintain authenticated users.
- Separation of frontend and backend code for maintainability.
- Creation of an administration panel for product management.
- Use of GitHub for version control and project delivery.

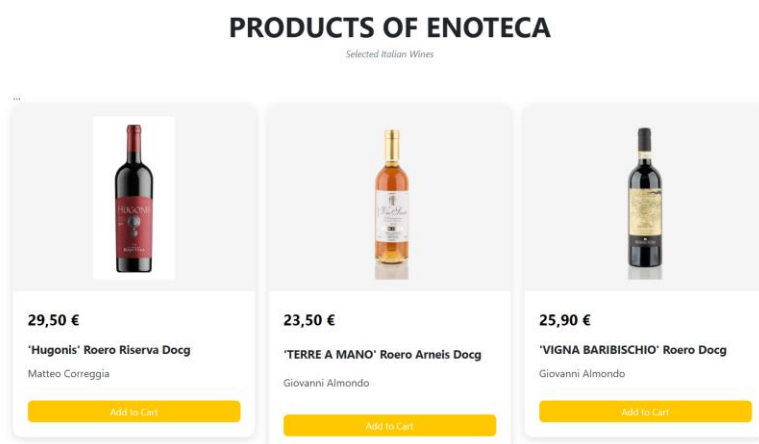
The graphical interface was designed to resemble a modern online Enoteca, emphasizing wine presentation through product images and clear navigation.

9. Screenshots

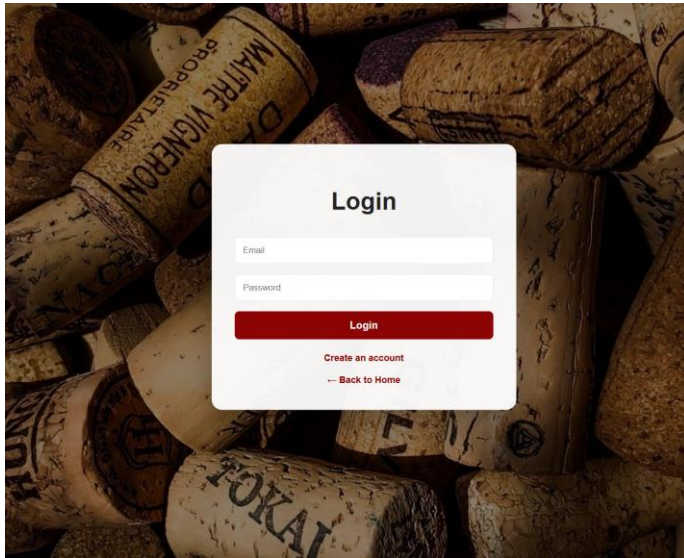
1. Home Page



2. Product Catalog



3. Login Page



Login

Email

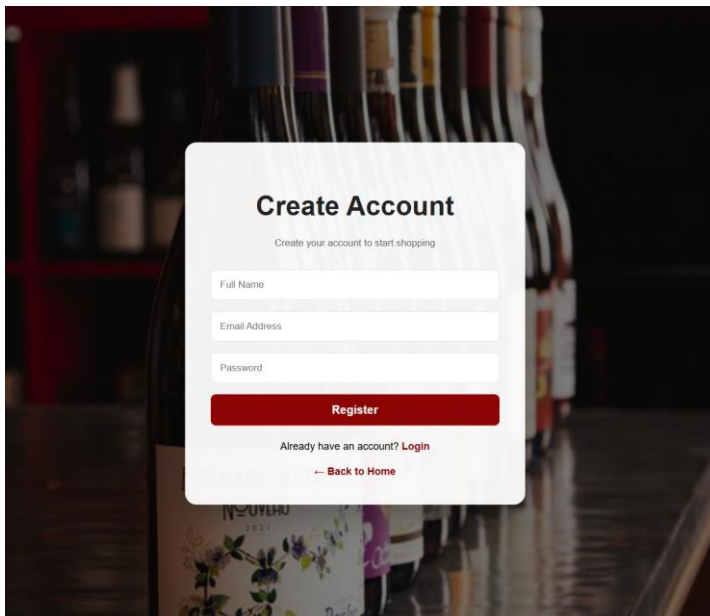
Password

Login

[Create an account](#)

[← Back to Home](#)

4. Registration Page



Create Account

Create your account to start shopping

Full Name

Email Address

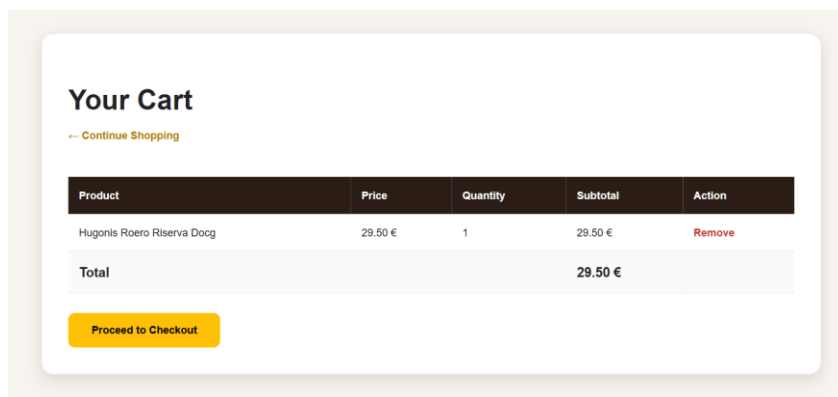
Password

Register

Already have an account? [Login](#)

[← Back to Home](#)

5. Shopping Cart



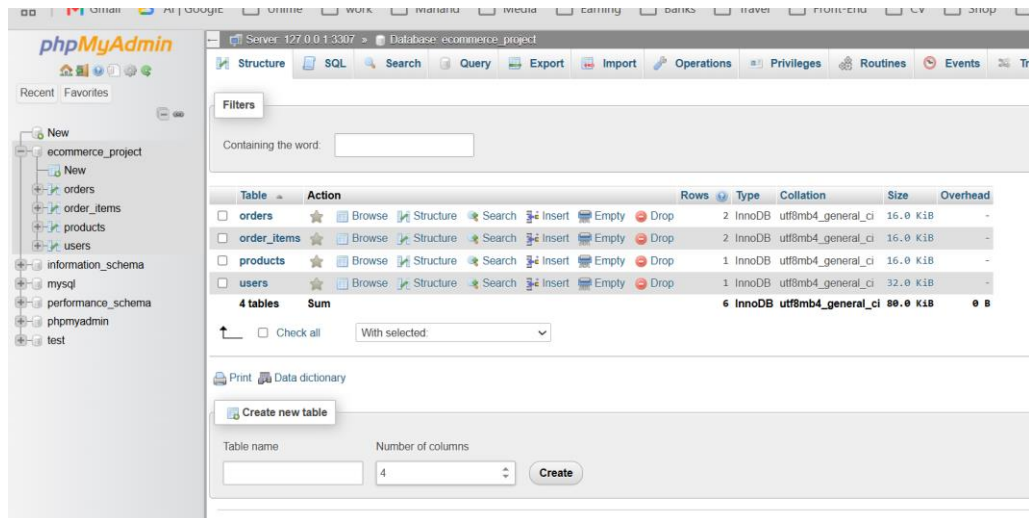
Your Cart

[← Continue Shopping](#)

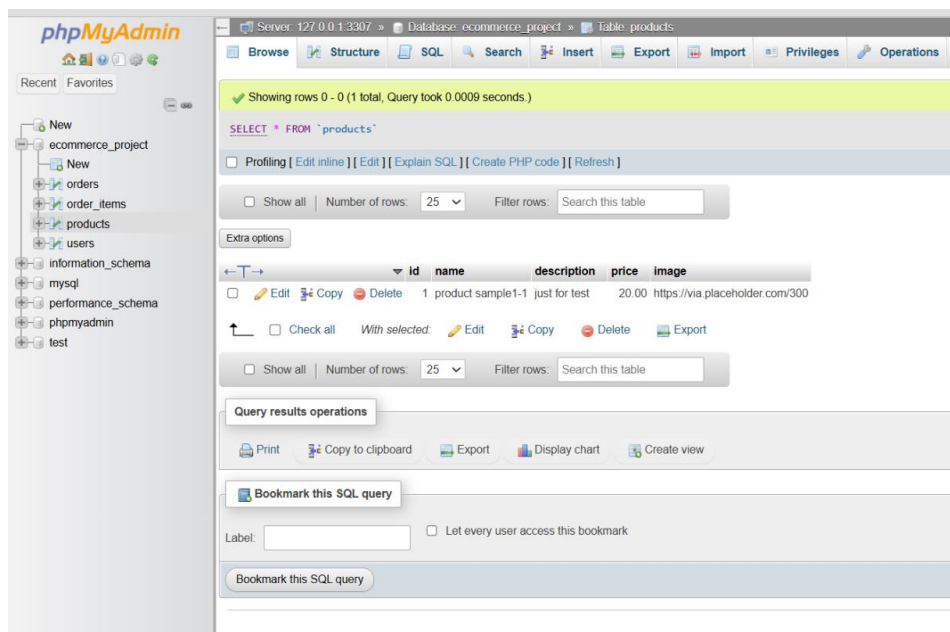
Product	Price	Quantity	Subtotal	Action
Hugonis Roero Riserva Doog	29.50 €	1	29.50 €	Remove
Total			29.50 €	

Proceed to Checkout

6. Admin Dashboard



7. Add/ Edit Product Page



10. Conclusion

This project successfully implements a complete e-commerce platform for an online Enoteca. The system allows users to register, authenticate, browse products, manage a shopping cart, and interact with a database-driven application.

The project demonstrates the practical application of HTML, CSS, JavaScript, PHP, MySQL, Git, and GitHub while satisfying all requirements specified in the Web Programming course guidelines.

Future improvements could include payment gateway integration, order history management, product categories, search functionality, and enhanced security features.